Pitot Heating (Mod 6/2033) – General Data

1. General

The pitot heads, have integral electrically heated elements which may be switched on to prevent ice formation which could block the pitot head.

2. Description

The aircraft is fitted with two pitot heads, one is installed on a mast projecting from the left side of the nose section, and a second is installed in an identical position on the right side of the nose section (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data). Pitot heat is controlled by a switch labelled PITOT HEAT, located on the LH sub panel. Each heating circuit is monitored by current sensing relays which provide grounds to the MAU when the current flow is interrupted.

28 Volt DC power is provided from the left and right buses through a 7 1/2 amp PITOT HEAT L (A6) circuit breaker and a 7 1/2 amp PITOT HEAT R (B6) circuit breaker on the main circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

Two configurations are available for displaying the Left and/or Right Probe Fail, depending on aircraft operational requirements. The configuration is selected by changing the DAU configuration plug (refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data). The configurations are outlined below:

A. Configuration 1 – A CAS message is displayed whenever the current flow to a pitot head is interrupted. This includes when the PITOT HEAT switch is selected to the OFF position.

B. Configuration 2 – A CAS message is only displayed when the PITOT HEAT switch is selected to the ON position and current flow to the pitot head is interrupted.

3. Adjustment/Test

A. Operational Test

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Ensure PITOT HEAT L (A6) and PITOT HEAT R (B6) circuit breakers are closed.

3) Select PITOT HEAT switch to ON and check that pitot head warms up.

CAUTION DO NOT LEAVE PITOT HEATER SWITCHED ON FOR MORE THAN A MINUTE.

4) Confirm no Left and/or Right Probe Fail CAS messages are present.

5) Pull the PITOT HEAT L (A6) circuit breaker and confirm the Left Probe Fail caution CAS appears.

6) Pull the PITOT HEAT R (B6) circuit breaker and confirm the Right Probe Fail caution CAS appears.

7) Reset both circuit breakers and confirm the CAS messages clear.

8) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. Removal and Installation

A. Pitot Heads

Removal/Installation of the pitot heads is detailed in 34-11-00, Pitot-Static System (Mod 6/2033) – General Data.

B. Pitot Heat

1) Removal of Pitot Heat Relay

a) Ensure the aircraft electrical system is OFF.

b) Open the PITOT HEAT L (A6) and PITOT HEAT R (B6) circuit breakers and fit safety clips.

c) Gain access to the fwd bulkhead by removing the MFDs as required (refer to 31-13-00, Apex Controllers – General data).

d) Remove (4) electrical terminals. Protect terminals from short circuits to airframe or each other.

e) While supporting the relay, remove (2) mounting screws.

f) Remove unit from bulkhead.

2) Installation of Pitot Heat Relay

a) Ensure the aircraft electrical system is OFF.

b) Set relay in position and install (2) mounting screws.

c) Remove protection and install (4) electrical terminals. Ensure terminals are tight.

d) Install MFDs as required (refer to 31-13-00, Apex Controllers – General data).

e) Remove safety clips and close the PITOT HEAT L (A6) and PITOT HEAT R (B6) circuit breakers.

5. Troubleshooting

A. Pitot Heat

1) Ensure PITOT HEAT L (A6) and PITOT HEAT R (B6) circuit breakers are closed.

2) Visually inspect the pitot heat connectors for damaged wiring or contacts. Ensure electrical connectors are fully seated.

3) Visually inspect the electrical terminals on the pitot fail relays for damaged wiring. Ensure the nuts holding the electrical terminals are tight.

4) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

5) If repair or service is required, send to appropriately rated avionics repair facility.